

CSCI 5942, FALL 2026

AI Engineering

Models, data, and the systems that train and serve them

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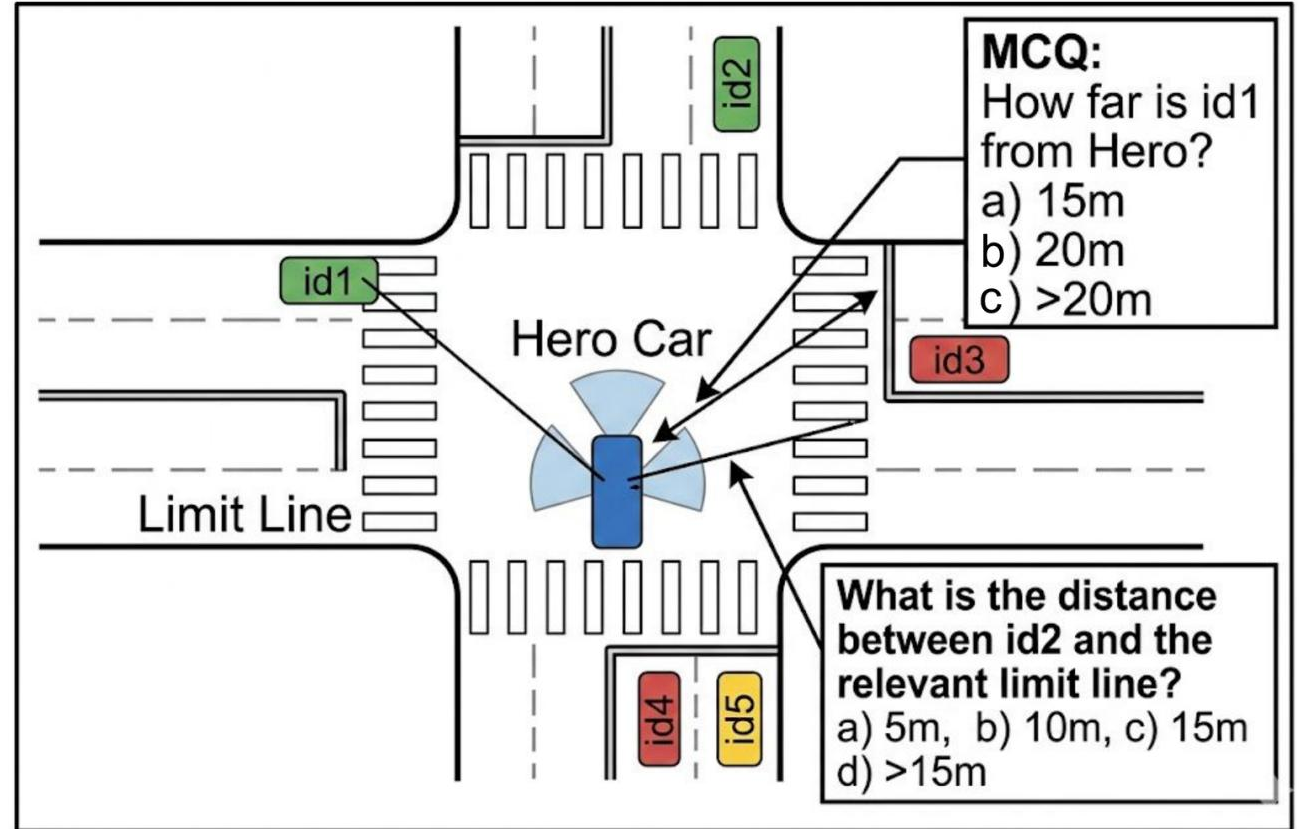
No Peeking

MCQ

What is the distance between id2 and the relevant limit line?

- a) 5m
- b) 10m
- c) 15m
- d) >15m

You need the scene. So here it is.



A fine-tuned VLM scores +67% over random on questions like these with the video switched off.

Chris Heckman

Associate Professor of Computer Science, and Director of the **Autonomous Robotics and Perception Group**

Programs

NSF CAREER · visually degraded environments
DARPA Subterranean Challenge · MARBLE, third place
USDA-NIFA NRI · revegetation robotics
DARPA AIDA · RAMFIS, intelligence community
ARL DCIST · distributed collaborative autonomy

Industry

Amazon Scout · sidewalk delivery
Zoox · autonomous driving, senior staff



The Lab



DARPA SubT, Team MARBLE

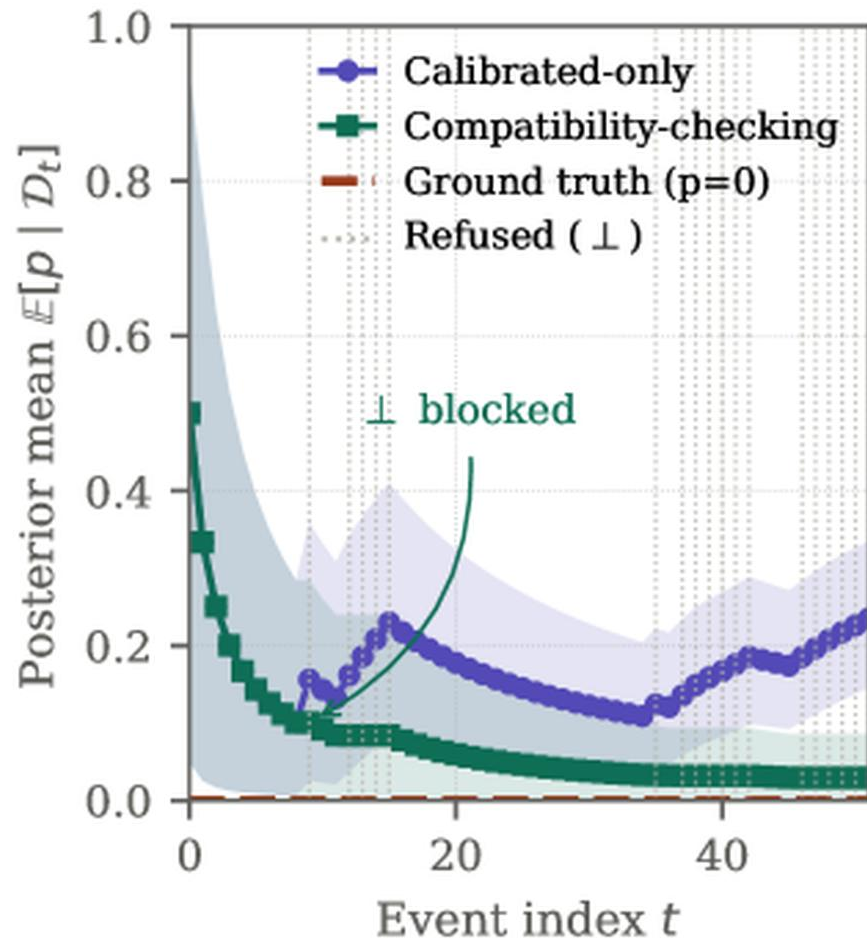


USCGC Healy, with CRREL and UAF



Canyonlands Research Center

Belief Consistency



Two channels, one map

geometry, with an error model we trust
a foundation model, confident and uncalibrated

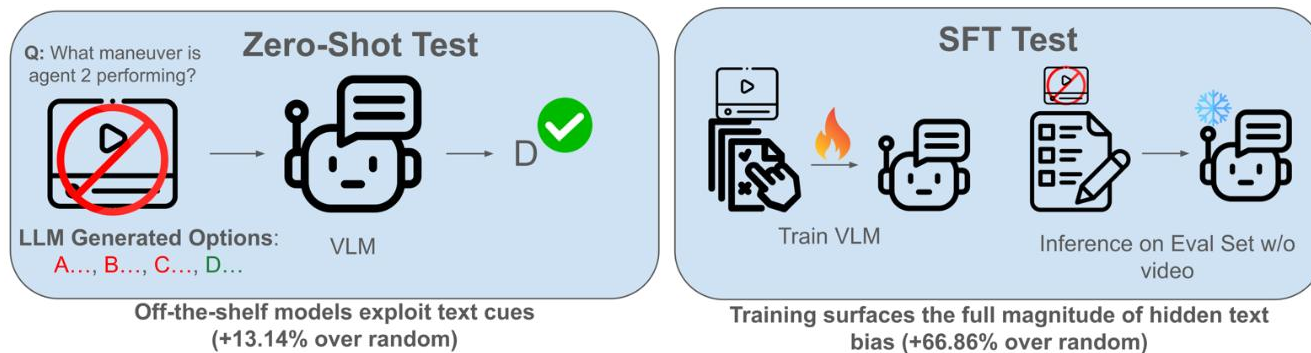
The operator

per-class calibrated commit gate
conflict-drop window at the moment of the claim

KITTI-360, is_car

commit precision 99.6% against 44.0%
mean per-class IoU 0.485 against 0.205

How VLMs Cheat

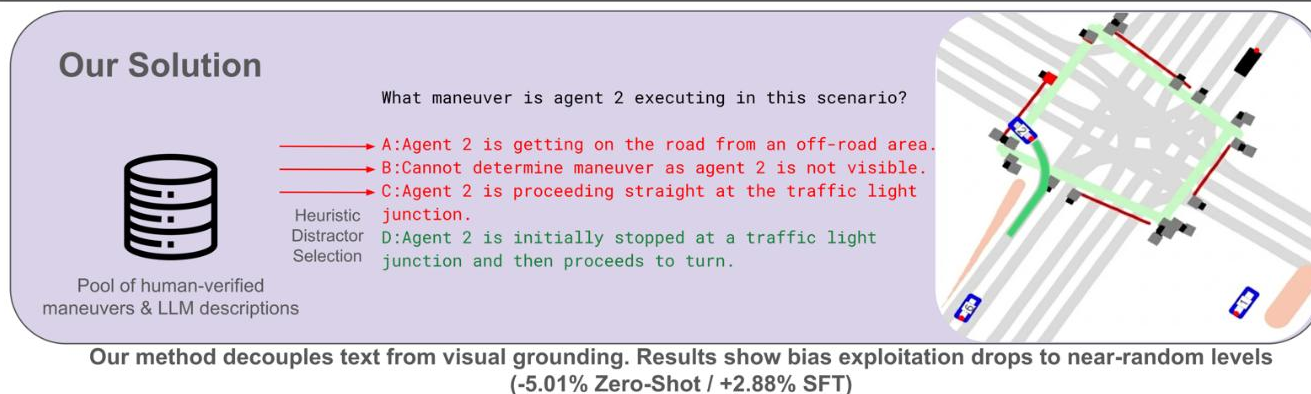


The setup

synthetic driving questions, written by an LLM
we blocked the video and asked anyway

What came out

+13% over random, off the shelf
+67% after fine-tuning, with no video at all

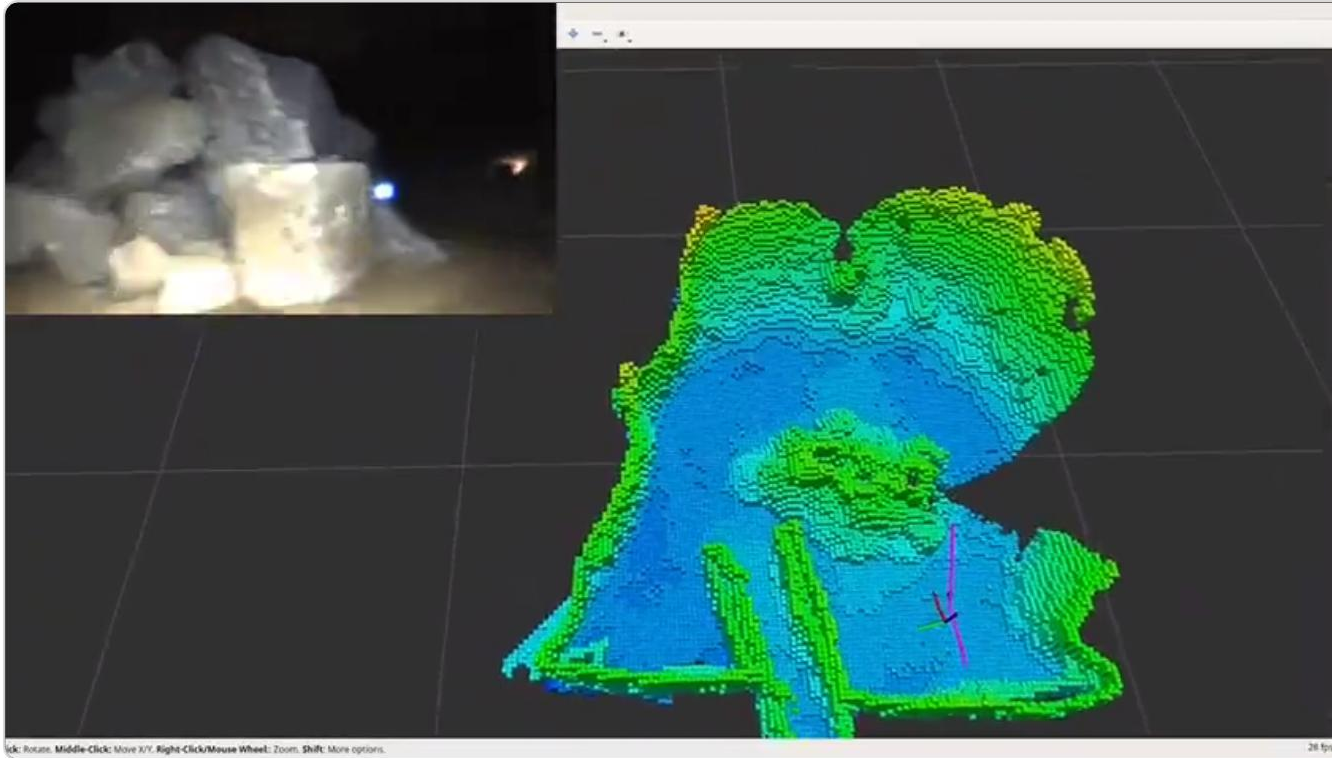


The fix

heuristic distractor selection
back down to +2.9%

Kulgod, Ye, Tanwar, Heckman. *Reducing Text Bias in Synthetic MCQs for Driving VLMs*. arXiv:2602.17677, with Zoox.

The Embodiment Gap



Spot at the DARPA SubT finals, routing on its own map

Six orders of magnitude

text corpora, 10^{13} tokens

robot trajectories, 10^7

You cannot scrape the rest

the clip is one route, one robot, one day

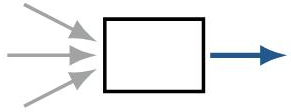
every trajectory is paid for in field time

So the frontier is data

simulation as a data factory

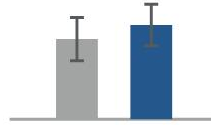
pooling across labs, Open X-Embodiment

What Transfers Back



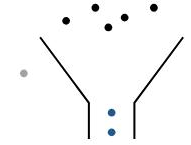
Serving under a deadline

the loop closes at 10 Hz
slow backbone, fast head
quantization on the robot



Evaluation you cannot fake

no A/B test on a robot
a wrong claim gets committed



Pooling across labs

Open X-Embodiment
sim as a data factory

The Split

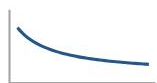
Models (me)

architectures, scaling
data
fine-tuning, post-training
evaluation
applications

Systems (Mark)

MLOps, data infrastructure
distributed training
inference serving
agentic systems
sustainability

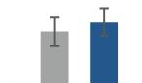
By December, on infrastructure you configured:



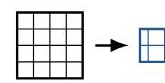
trained



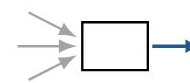
adapted



evaluated

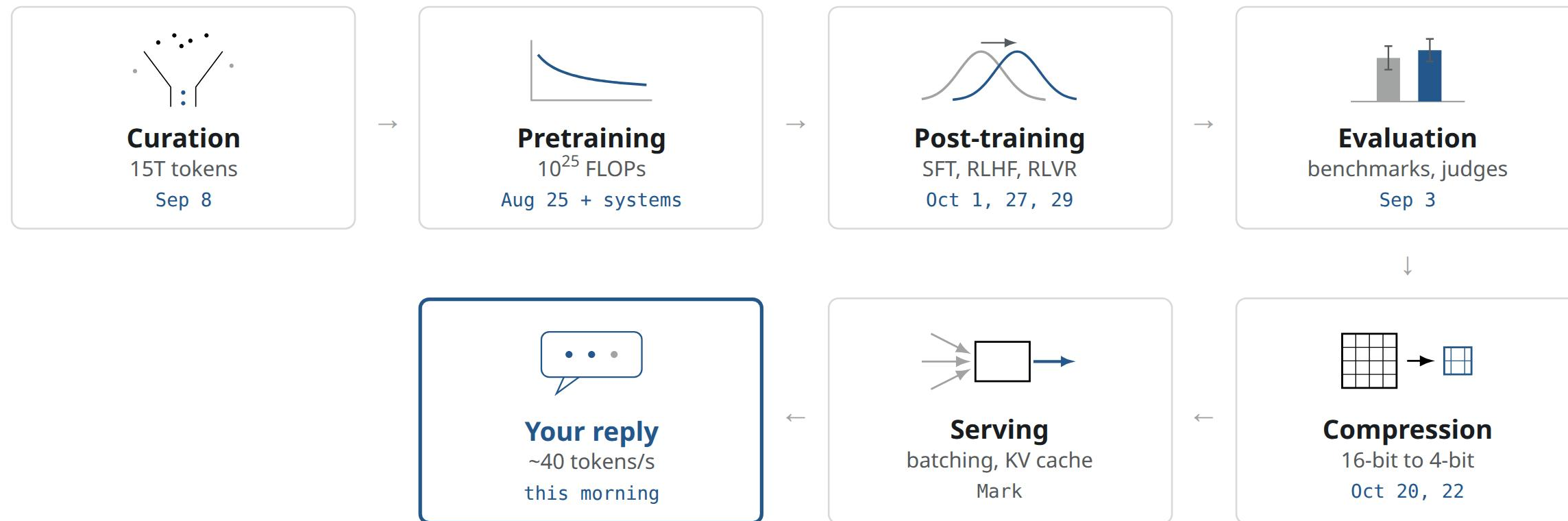


quantized

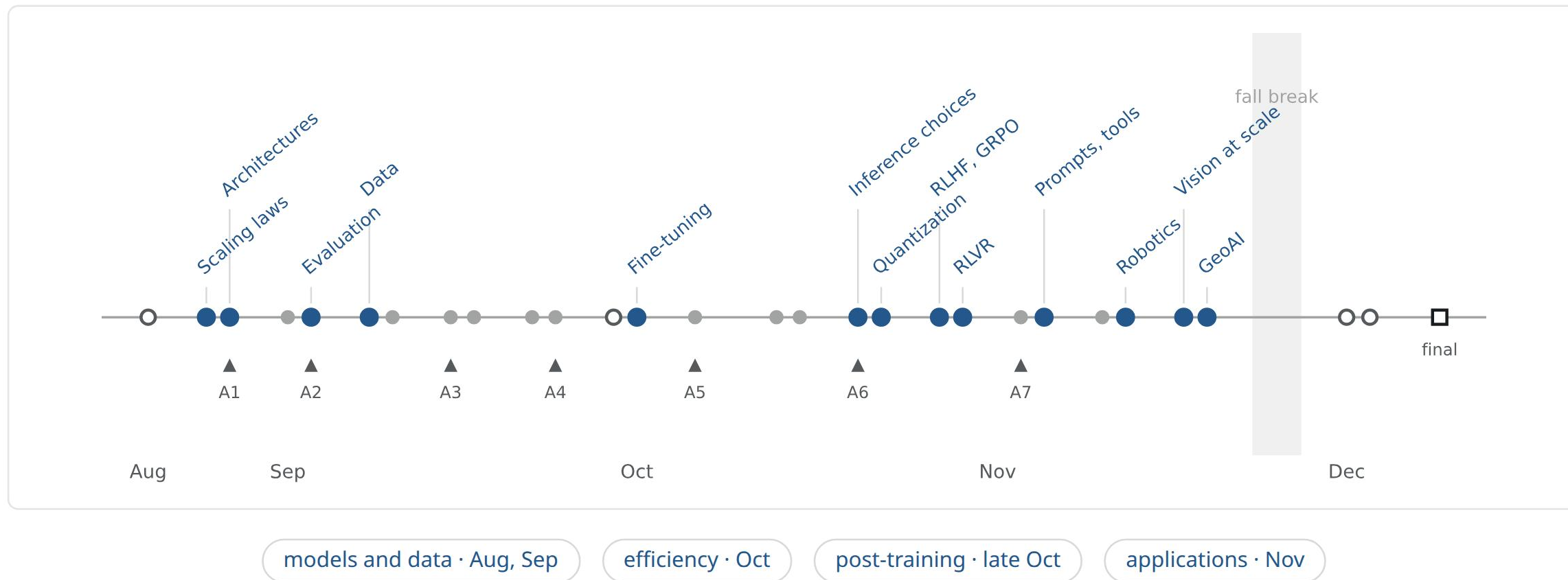


served

Behind One Reply



My Thirteen Lectures



How the Course Runs



Assignments · 50

seven builds, each with an in-class quiz

Mega assignment · 20

group build, poster session Dec 3

Participation · 10

Socratic round robin

Final exam · 20

written, in person, Dec 9

The Fine Print

Late work. Two days late on any assignment, no penalty. After that, 25% per additional day. You get **five late days** for the whole semester, counted automatically from submission time.

AI tools. Permitted, and treated as if the AI were a colleague: consult it, delegate to it, learn from it. You own the result, including its correctness and your understanding of it. On the mega assignment AI use is expected, so document it as you would any collaborator.

Quizzes and the final. Closed book. No devices, no AI.

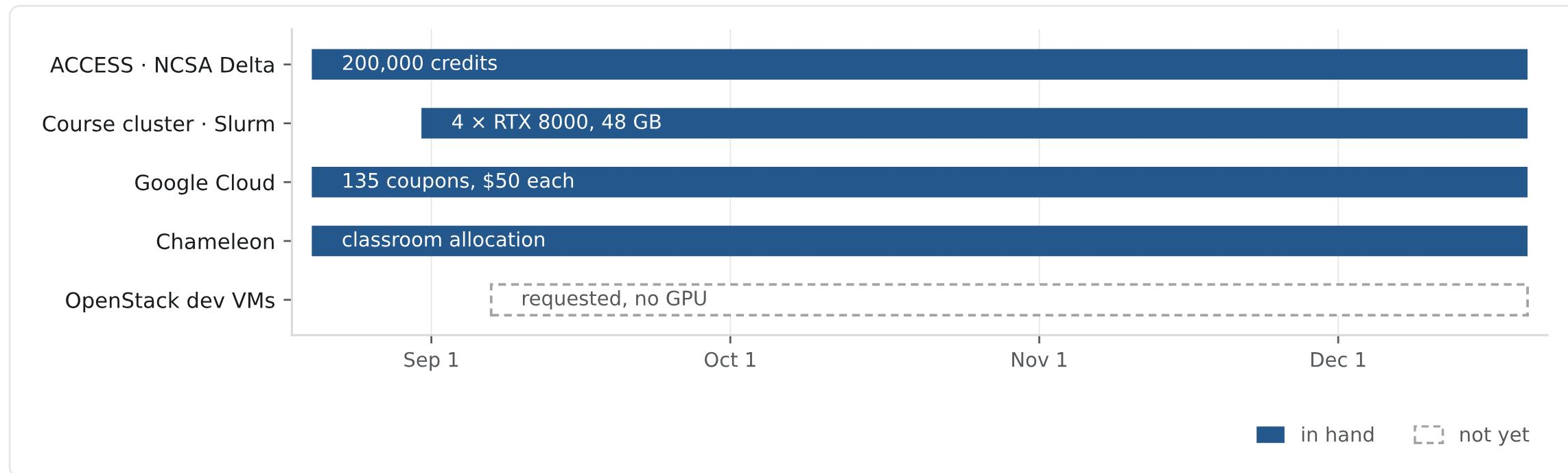
Collaboration. Study groups are encouraged. Everyone submits their own work and lists their study-group members at the top.

Where things live. This website is the source of truth for schedule and readings. Assignments and starter code are in the GitHub org, submission is Gradescope, announcements and discussion are on Slack.

Prerequisites. CSCI 4622 and CSCI 4922.

Your TA. Nolan Brady, Nolan.Brady@colorado.edu.

Compute



Today

A0: environment setup

Google Cloud signup, from Mark's link
one coupon per colorado.edu address

Where the work runs

ACCESS credits, the main pool

course cluster from end of August

Chameleon and dev VMs if we need them